

Theoretical risks and tabular asterisks: Sir Karl, Sir Ronald, and the slow progress of soft psychology

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Abstract

Theories in “soft” areas of psychology lack the cumulative character of scientific knowledge. They tend neither to be refuted nor corroborated, but instead merely fade away as people lose interest. Even though intrinsic subject matter difficulties (20 listed) contribute to this, the excessive reliance on significance testing is partly responsible, being a poor way of doing science. Karl Popper’s approach, with modifications, would be prophylactic. Since the null hypothesis is quasi-always false, tables summarizing research in terms of patterns of “significant differences” are little more than complex, causally uninterpretable outcomes of statistical power functions. Multiple paths to estimating numerical point values (“consistency tests”) are better, even if approximate with rough tolerances; and lacking this, ranges, orderings, second-order differences, curve peaks and valleys, and function forms should be used. Such methods are usual in developed sciences that seldom report statistical significance. Consistency tests of a conjectural taxometric model yielded 94% success with zero false negatives.

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✕ Deceased.